

SUNSPOKE · PROBLEM STATEMENT

SunSpoke problem statement

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-24	Amish Chadha	Populate to TRL 2 (users, context, constraints, out of scope, prior work)
0.1	2026-09-24	Amish Chadha	Initial scaffold

SunSpoke problem statement

Most bicycles in rural sub-Saharan Africa are unpowered steel roadsters that carry people, water, firewood and goods over long, rough, hilly routes, and the electric alternatives on sale are either a whole new e-bike or a closed kit that local mechanics cannot repair and that assumes grid charging. Design with, not for: requirements must come from co-design sessions and field trials with the intended users through a local partner.

■ The problem

The heavy single-speed roadster (28 in wheels, steel frame, rod or cable brakes, strong rear carrier) is the working vehicle of rural East, Central and Southern Africa and of many low-income regions elsewhere. Riders routinely carry 30 to 80 kg of goods or a passenger over 10 to 30 km a day on dirt roads. The limit is the rider's own power: hills and loads turn a 20 km trip into hours of walking and pushing.

Electric assist solves this, and e-bikes and conversion kits are already sold in the region. Three gaps remain:

1. **Fit.** Most kits are designed for modern aluminium bikes with 26 or 29 in wheels, disc brakes and threaded bottle-cage bosses. Roadsters have 28 in (ETRTO 635) rims, narrow steel dropouts, rod brakes and no mounting bosses. Buying a new e-bike to get assist throws away a bike the owner already has.
2. **Repair.** Closed kits use proprietary connectors, sealed displays and controller settings that need a laptop or app. When a part fails, the bike is parked until a spare arrives from a city. The roadside bicycle mechanic, who keeps every other bike running, is locked out.
3. **Charging.** Grid power is absent or unreliable in much of the rural region. Charging at a shop in town costs money and a trip. A single 100 W panel or a shared village charging hub could supply enough energy, but kits are not designed around slow solar charging.

SunSpoke is an open, bolt-on conversion kit that fits the roadsters people already own, can be fitted and repaired by a local bike mechanic with basic tools, and charges from one 100 W panel or a shared hub using a SwapCell-compatible pack.

■ Users and context

User	Need	Context
Rural commuter	Reach work, school or market 10 to 30 km away without arriving exhausted	Dirt and gravel roads, hills, daily use
Small trader	Carry 30 to 80 kg of produce, water or goods to market, sometimes a passenger	Loaded rear carrier, stop-start riding, heat
Community health worker	Reliable daily rounds to scattered households with a medical kit	Long distances, schedule matters, often alone
Local bike mechanic	Fit, maintain and repair the kit with the tools and parts already in the shop, and earn from it	Roadside or market stall, hand tools, no welding in many cases
Village charging hub operator	Charge several packs a day from a shared solar array and track their condition	Hub with panels, dock and shade

Operating environment

- **Roads:** murram (laterite), gravel, sand and rutted dirt, with potholes, washboard and grades up to about 10 % on short climbs.
- **Loads:** rider 60 to 80 kg plus 30 to 80 kg on the carrier; bike about 20 to 23 kg before conversion.
- **Climate:** ambient 15 to 40 °C, with surfaces in sun reaching 60 °C or more; dust in the dry season; heavy rain, mud and shallow water crossings in the wet season.
- **Power:** limited or no grid; solar resource of about 4 to 6 peak sun hours per day across most of the region, lower in the rains.
- **Supply chain:** bicycle spares, generic e-bike parts and motorcycle electrical parts are available in towns; specialist parts are not.

■ Constraints

- Garage-buildable prototype, about \$250 USD for the kit. The SwapCell pack is costed separately (see `bom/bom-notes.md`).
- Bolt on to the donor bike with no welding, frame drilling or cutting.
- Fitting and repair possible with the tools a roadside bike mechanic already has.
- Standard, pluggable connectors and generic parts wherever possible, so a failed part can be swapped with one from the local market.
- Legal and safe assist: 250 W rated power and an assist cutoff speed in line with pedelec practice (EN 15194 sets 250 W and 25 km/h in Europe); national rules in target countries to be checked.
- Uses the SwapCell pack interface so packs can be shared across portfolio vehicles and charged at SwapCell docks. The motor voltage decision that follows from this is open (see SSP-PRC-001).

■ Out of scope

- Building or selling complete bicycles.
- Designing the battery pack itself (that is SwapCell).
- The village charging hub and its dock (SwapCell and PowerBox cover these).
- Motorcycle-class speed or power (over 250 W rated or assist above 25 km/h).
- Cargo tricycles and trailers (see CargoMule and FlatTrike).

■ Prior work

- **Commercial conversion kits.** Front hub, rear hub and mid-drive kits with 36 V and 48 V batteries are sold worldwide and in regional markets. They prove the technology and set the price floor, but are not designed for 28 in roadster wheels, rod brakes or local repair.
- **Regional e-bike ventures.** Several companies in East Africa assemble or sell e-bikes and electric motorcycles, some with battery swapping. They show demand and a working swap model, but most are closed products that need their own service network.
- **Rugged bicycles for rural use.** Purpose-built heavy-duty bicycles such as the World Bicycle Relief Buffalo bicycle show that design for rural loads, local assembly and mechanic training works at scale.

- **Bicycle taxis (boda-boda).** Bicycle taxis in East Africa show how much load and distance a roadster is asked to carry, and how important a local repair economy is.
- **Solar charging for small vehicles.** Solar e-bike charging stations and off-grid battery hubs exist in pilot form. The first-order numbers in SSP-PRC-001 show one 100 W panel can cover a typical day of assisted riding.

Links and named sources will be added at TRL 3 once each is checked.

■ Open questions

- Which partner organization and which region first (for example an East African bike mechanic cooperative, a health worker program, or a university engineering department)? Proposed, awaiting Amish.
- Is a 48 V system acceptable locally, or are 36 V spares so much more common that a 36 V kit is preferred? This drives the motor voltage decision in SSP-PRC-001.
- How much cargo and passenger carrying must the kit support, and does the pack location need to leave the rear carrier free? Assumed yes; to validate.
- What assist speed limit and registration rules apply in the first target country?

■ User research and co-design

This design is for communities the author is not part of, so requirements come from the people who will use it.

- [] Identify a local partner organization (Helpful Engineering network, NGO or university)
- [] Run co-design sessions with intended users; record who, where and what was learned
- [] Validate load, distance, terrain and cost assumptions in the field
- [] Revise requirements (REQ) from findings before freezing the design